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Camera-relocalization-driven monocular 3D safety monitoring system for crane operations☆

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ABSTRACT

Construction remains one of the most hazardous industries, with tower crane operations posing significant risks under limited visibility and dynamic site conditions. This paper proposes a calibration-free vision-based framework for safety-oriented 3D localization using crane-mounted monocular cameras. The D2S-MDE framework recovers metric-scale 3D worker positions from a single RGB image by enforcing geometric consistency between camera relocalization and monocular depth estimation. Sparse metric depth anchors from scene coordinate regression are used to align dense monocular depth, enabling robust metric depth reconstruction under dynamic viewpoints. The estimated 3D positions are integrated into a BIM-aligned digital twin in NVIDIA Omniverse for real-time safety monitoring and risk assessment. Experiments demonstrate decimeter-level camera pose accuracy (0.012 m/0.019°) in real scenes and achieve a Height-Normalized Localization Score of up to 92.67% at 14 m, with risk assessment accuracy reaching 94.19%. This performance highlights the framework effectiveness for practical deployment in real construction environments.

1. Introduction

Construction is one of the most hazardous industries, accounting for a disproportionate share of occupational injuries and fatalities [1]. According to the statistics from the Bureau of Labor Statistics (BLS), the fatal injuries in construction sectors in 2023 were 1075 cases, leading to the top fatal rate among industries. Beyond the tragic human losses, such accidents also lead to economic burdens. For example, the Occupational Safety and Health Administration (OSHA) estimated that the workplace injuries cost almost $1 billion per week in 2023 [2]. These statistics underscore the critical need for effective safety management and proactive hazard monitoring on construction sites.

In the construction areas, the safety of workers working around heavy constructions machines, specifically tower crane, is particularly critical because these machines have very large sizes and their operators normally have limited visibility. Therefore, it would be a thorny problem for people to avoid potential hazards from these kinds of machinery while maintaining the continuous operations, especially in a crowded and dynamic environment. As a result, developing an advanced monitoring system for real-time visualization and safety risk awareness should be taken into account as priorities.

In the past decade, vision-based hazard monitoring has gained significant attention due to advances in computer vision. Early studies primarily focused on 2D worker and equipment detection for construction safety monitoring using CCTV and crane-view imagery. For example, Price et al. [3] developed a multisensor crane monitoring system for blind lift operations, while Sutjaritvorakul et al. [4] proposed a data-driven worker detection framework using crane-view cameras. Recent studies further addressed challenging construction scenarios such as overlapping worker detection [5], long-tailed object detection under domain-shifted environments [6], and multiview trajectory analysis for construction-object monitoring [7]. Other studies extended safety monitoring to the 3D level. For example, Kulinan et al. [8] estimated 3D worker coordinates within a BIM platform, while Lee et al. [9] employed stereo vision for multi-worker localization and tracking. To support dynamic camera systems, Wang et al. [10] combined visual odometry and monocular depth estimation for crane monitoring, whereas Fu et al. [11] proposed a 2D–3D matching pipeline for crane boom-end localization. Han et al. [12] further integrated UAV monitoring with digital twin environments for large-scale worker management. In parallel, BIM- and digital-twin-based safety management has attracted increasing attention. Tak et al. [13] and Shahnavaz

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et al. [14] simulated crane operations within BIM environments for clash detection and motion-path analysis, while Kulinan et al. [15] extended BIM integration toward dynamic hazard analysis using CCTV data.

Despite the remarkable progress in advancing safety monitoring technologies, several limitations persist. Firstly, there is fragility and computational inefficiency in image-reference-based 2D–2D and 2D–3D matching. A reference-based method retrieves candidate reference frames from the entire video, establishes 2D–2D correspondences, and then lifts inliers to 2D–3D using a prebuilt SfM point cloud followed by PnP. Construction-site imagery often contains repetitive structural textures (rebar and scaffolding), environmental disturbances (dust or rain), and motion-induced blur, which can degrade the reliability of 2D–2D feature matching. Moreover, pipelines that perform 2D–2D matching prior to 2D–3D lifting typically incur substantial computational overhead, posing challenges for real-time edge deployment. In this paper, we suggest bypass reference-frame retrieval by directly matching the 2D query image to a pre-stored 3D point cloud, achieving more noise-robust localization with markedly lower computational overhead. Beyond matching-based localization, recovering reliable depth information from monocular cameras remains another challenge. Existing monocular depth estimation methods often depend on externally provided camera poses and heuristic pixel filtering strategies, which can lead to inaccurate depth prediction and scale ambiguity. As a consequence, object-level 3D localization becomes unstable when 2D detections are projected using mismatched scaled depth values. Although sparse 3D descriptors are sometimes reconstructed in these pipelines, they are rarely exploited as geometric constraints to regularize dense depth estimation. In addition, most existing 3D localization paradigms are built upon multi-view or stereo vision assumptions, where depth is inferred from synchronized cameras or temporally adjacent frames. While effective for static surveillance systems, these assumptions break down in crane-mounted or mobile camera scenarios, where fixed stereo baselines are unavailable and temporal multi-view reconstruction introduces latency and instability due to insufficient viewpoint variation between consecutive frames. In parallel, existing multi-object 3D safety monitoring systems typically visualize workers, machinery, and safety zones independently, without explicitly modeling their spatial interactions within a unified environment. Furthermore, although digital twin technologies have been widely applied in civil engineering applications such as Structural Health Monitoring (SHM) [16,17], few studies integrate real-time 3D localization with interactive digital twin environments capable of rule-based hazard evaluation and spatial risk visualization.

As a result, 3D localization for dynamic construction cameras must be reformulated as a per-frame inference problem that jointly reasons about camera pose and scene depth from a single RGB image. However, this reformulation introduces two fundamental challenges: (i) camera relocalization must be lightweight and robust without relying on expensive point-cloud-based matching pipelines, and (ii) depth information must be inferred in metric scale despite the inherent ambiguity of monocular depth estimation. Addressing these challenges is essential for enabling real-time, calibration-free 3D safety monitoring with dynamically mounted cameras.

Unlike conventional multi-view or calibration-based pipelines, crane-mounted safety monitoring requires per-frame metric 3D localization under continuously changing viewpoints. In this setting, neither fixed stereo geometry nor repeated manual calibration is practical. To address this challenge, we formulate crane-mounted safety monitoring as a geometric consistency problem between scene coordinate regression and monocular depth estimation. Specifically, scene coordinate regression provides sparse metric geometric anchors through camera relocalization, whereas monocular depth estimation provides dense but scale-ambiguous depth predictions. By reconciling these two representations at inference time, the proposed framework enables metric 3D localization from a single moving crane-mounted camera without explicit camera calibration. We propose a calibration-free deep-learning framework that enables 3D localization from a single crane-mounted RGB camera by enforcing coordinate consistency at inference time, as summarized in Table 1. The key contributions of this study are summarized as follows:

- We introduce a calibration-free vision framework for crane-mounted cameras that enables metric 3D worker localization under dynamically changing viewpoints without requiring manual camera calibration or additional sensing hardware.
- We propose a scale-consistent depth recovery strategy that reconciles dense monocular depth predictions with sparse metric depth anchors derived from camera relocalization. This approach provides a practical mechanism to recover metric-scale depth under dynamic camera conditions commonly encountered on construction sites.
- We validate the proposed framework through experiments on real construction site data and demonstrate its integration with a digital-twin environment for rule-based safety monitoring and hazard evaluation.

2. Related work

This section reviews previous studies on camera relocalization, 3D vision-based localization, and digital-twin-based safety monitoring in construction environments. Existing approaches and their limitations are discussed to position the proposed framework within current research trends.

2.1. Camera relocalization on safety monitoring

Camera relocalization is the task of estimating a camera's 6-DoF pose from a single RGB image and can generally be categorized into three groups. The first category is image-retrieval-based localization, where query images are matched against geo-referenced image databases using global descriptors or place-recognition methods [19,20]. Subsequent studies improved retrieval robustness using synthesized views and learned global representations [21,22]. The retrieved reference images are then used to approximate the camera pose [23]. Although this approach is computationally efficient, it depends heavily on dense reference-image coverage and often struggles under unseen viewpoints or environmental changes. Structure-based approaches instead estimate camera poses through explicit 2D–3D correspondences between query images and reconstructed scene geometry. Representative methods include feature metric localization [24], pixel-level pose refinement [25], hierarchical localization [26], and graph-based feature matching frameworks such as SuperGlue and SuperPoint [27,28]. These methods generally achieve high localization accuracy by leveraging geometric constraints, but they require dense 3D maps and computationally expensive feature matching during inference. To address these limitations, end-to-end deep learning methods based on Scene Coordinate Regression (SCR) have been introduced. Early SCR approaches employed lightweight regression networks to directly predict scene coordinates from RGB images [29,30], while more recent studies improved robustness and scalability for large-scale environments [31,32]. Since SCR-based methods estimate camera poses directly from predicted scene coordinates without explicit image retrieval or dense point-cloud matching, they provide a lightweight and practical solution for dynamic-camera scenarios in hazard monitoring.

Camera relocalization has been widely applied in robotics and autonomous vehicles for navigation. Recognizing its potential, researchers have also explored its application in heavy construction equipment, particularly tower cranes. One early approach is proposed by Jing et al. [33], who align live camera frames with a point cloud map to approximate the crane camera pose. They generate a wide-angle synthetic reference image from the point cloud and employ 2D feature

Table 1
Comparison of current studies on safety monitoring using camera calibration.

Research	Output	Sensors	Approach	Auto Relocalization	Risk Assessment	DT Integration
Chian *et al* [18]	2D safety zone, worker positions	Static CCTV	Manual 2D spatial mapping and object detection for load fall zone calculation	x	o	x
Kulinan *et al* [8]	3D worker position	Static CCTV	Single-camera position estimation; BIM visualization	x	o	o
Lee *et al* [9]	3D worker tracks	Multiple CCTV	Stereo vision worker identification and tracking	x	x	x
Wang *et al* [10]	3D camera poses, WAL	Dynamic camera	Semi-direct Visual Odometry (SVO) and REMODE for spatial mapping	o	x	x
Ling *et al* [11]	3D camera poses	Dynamic camera	2D–3D matching; sparse point cloud localization	o	x	x
Han *et al* [12]	3D worker position	Drone GPS; camera	GPS-based localization; ID data matching; WHSC Drone Intelligent Inspection System	o	x	o
Ours	3D worker positions; dynamic safety zone	Dynamic camera	Scene Coordinate Regression (SCR) relocalizer; Depth Anything V2 for depth estimation; NVIDIA Omniverse visualization	o	o	o

matching (SIFT) with RANSAC to estimate a 2D transformation of the camera view. While this method is computationally efficient, it only provides 2D pose alignment and becomes less reliable when the camera moves, as the overlap with the static reference image decreases. Consequently, its accuracy drops during fast crane operations. Another approach for safety monitoring is Semi-Visual-Odometry-based relocalization, demonstrated by Wang et al. [10]. This method employs SVO for real-time pose estimation and geometric reconstruction, and then performs online relocalization of new images via 2D–3D SIFT feature matching. While the method achieves low errors in field tests, its performance is evaluated under relatively static conditions. Therefore, its adaptability to construction sites with frequent changes or moving objects remains uncertain. More recently, Fu et al. [11] introduce a structure-based technique to localize a crane boom tip using a sparse point cloud reconstructed via Structure-from-Motion. They employ image retrieval to enhance 2D–3D feature matching between crane images and the point cloud, and accurately determine the 3D position of the boom tip through geometric alignment. However, localization errors may increase when query images differ significantly from the reference set, resulting in insufficient feature correspondences between images and the sparse point cloud. As a result, the method requires highly overlapping images in the dataset to ensure full scene coverage and reliable localization.

2.2. 3D vision-based localization in safety monitoring

Compared with 2D visual detection, the use of 3D spatial information is widely acknowledged to better represent the interactions between workers and machinery on construction sites. The most common approaches introduce spatial localization frameworks based on stereo vision systems, among which Lee et al. [9], Cai et al. [34], and Zhang et al. [35] are representative examples. Generally, the method relies on the intersection view of at least two cameras simultaneously and includes four steps: camera calibration to estimate intrinsic and extrinsic parameters, 2D object localization to obtain image coordinates, object matching where visual information is associated across images based on epipolar geometry constraints, and triangulation which integrates the previous results to recover 3D coordinates of target entities. Despite their effectiveness, these methods still require significant manual calibration effort and accurate multi-view data association, which remains challenging in complex construction environments. In addition, existing stereo camera systems primarily focus on ground-level hazards rather than overhead risks, making them less suitable for crane operations.

To address this gap, the use of single camera, in collaboration with other sensors and AI-driven techniques, is a potential solution to estimate 3D entity coordinates from dynamic cameras. For instance, Zhang et al. [36] used a single monocular camera and geometry feeded from drone. After estimating the drone camera poses by Semi-Direct Visual Odometry, they combine adjacent frames with Multi-View Stereo method to generate depth map to ground plane, enabling back-projection of 2D detections into 3D coordinates based on known camera height and ground geometry. Meanwhile, Shen et al. [37] propose a direct approach, namely the Self-supervised Monocular Depth Estimation Network (SNDEU), which predicts both depth maps and unknown camera intrinsics from raw video sequences. The detected 2D results, including bounding boxes and instance segmentation masks, are projected into 3D space to form a pseudo-LiDAR point cloud, which is subsequently filtered using a density-based clustering algorithm to extract object-level 3D coordinates. Another method introduced by Han et al. [12] who links 2D detection results across UAV-captured video frames using multi-object tracking and maps them onto a coarse 3D photogrammetry model using UAV GPS data. However, unlike UAV systems equipped with accurate GPS, crane-mounted cameras lack predefined camera parameters, and their motion is often non-ideal, as intermittent stops during operation can negatively affect localization performance. Although depth estimation has shown promise for addressing these challenges [38], monocular depth predictions often suffer from scale ambiguity [39], particularly at long distances. Therefore, by incorporating sparse scene descriptors obtained from Scene Coordinate Regression (SCR), the proposed framework has the potential to overcome these limitations and provide more robust 3D localization in dynamic construction environments.

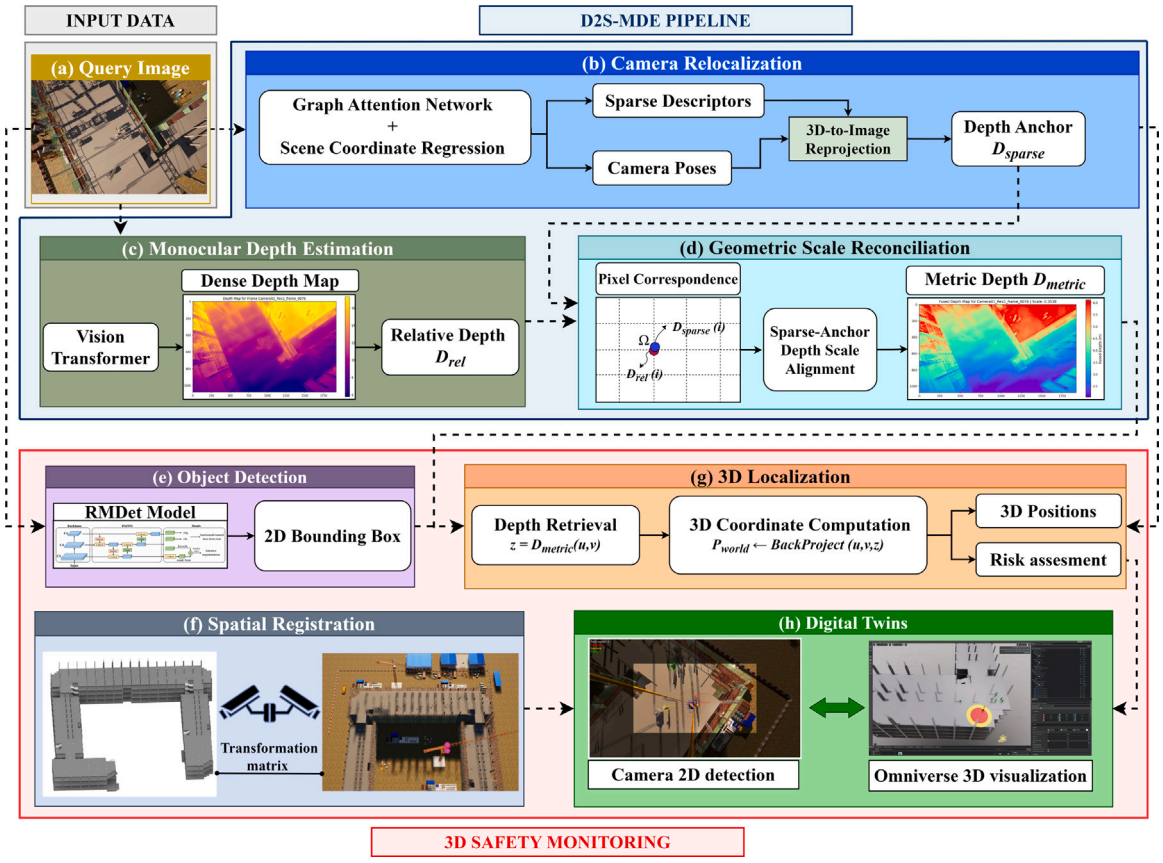


Fig. 1. Overall framework of D2S-MDE and 3D safety monitoring.

2.3. Digital twin-based hazards visualization

Digital Twins (DTs) are increasingly recognized as a bridge that integrates real-time data from sensors and other sources with management systems, enabling dynamic monitoring, risk assessment, and decision-making support. In construction safety management, DTs can be combined with computer vision and Building Information Modeling (BIM) to digitize risk assessment, facilitating real-time monitoring of workers' locations and safety compliance within a virtual environment [8]. Moreover, DTs provide an effective means not only for visualizing site operations but also for supporting task assignment and resource allocation [12,40].

DTs have significant potential to improve crane operation safety due to their ability to provide diverse types of data. Although it had been applied in certain cases, for example as analysis stability of tower crane hoisting, the most powerful feature is to provide the interaction throughout entities has not been properly utilized. To address this gap, we extend these concepts by integrating camera relocalization and monocular depth estimation outputs into a digital twin workspace developed in NVIDIA Omniverse, which has been shown effectively in progress monitoring [41] but in safety management. By mapping relocalized camera poses and depth-derived 3D coordinates of detected entities onto a virtual site model, the DT enables real-time visualization of spatial relationships among the crane boom, site workers, and hazardous zones. This spatial context is critical for assessing potential collision risks.

3. Methodology

This study proposes a vision-based framework for estimating the 3D positions of workers operating around crane zones by integrating camera relocalization and monocular depth estimation, as illustrated in Fig. 1. First, a top-down query image (a) is provided as input to the D2S camera relocalization module (b), which utilizes a graph attention network with scene coordinate regression to estimate the camera pose. Sparse scene descriptors obtained from this module are reprojected into the image plane to generate sparse depth anchors. In parallel, the same image is processed by a vision transformer-based monocular depth estimation module (c) to produce a dense relative depth map. The sparse depth anchors are then used to perform geometric scale reconciliation (d), aligning the relative depth map with metric scale to obtain a metric depth map. Subsequently, objects of interest are detected using the real-time object detection model RTMDet (e), which extracts 2D bounding boxes of workers in the image. By retrieving depth values from the metric depth map and applying back-projection using camera intrinsic parameters, the corresponding 3D positions of the detected entities are computed (g). Finally, the estimated 3D positions are spatially registered with the preloaded BIM model within the NVIDIA Omniverse environment (f). This enables digital twin-based visualization and real-time safety assessment for crane operations (h).

3.1. D2S-based camera relocalization

This section describes the proposed D2S-based camera relocalization framework for estimating crane-mounted camera poses from monocular images. The overall process includes data collection, scene coordinate regression training, and camera pose estimation using 2D–3D correspondences.

3.1.1. Data collection

To estimate new camera poses in a specific environment, a scene coordinate regression model requires two types of input data for training. First, 2D images with known camera intrinsic parameters and a sparse reconstruction point cloud are required. Top-down view images

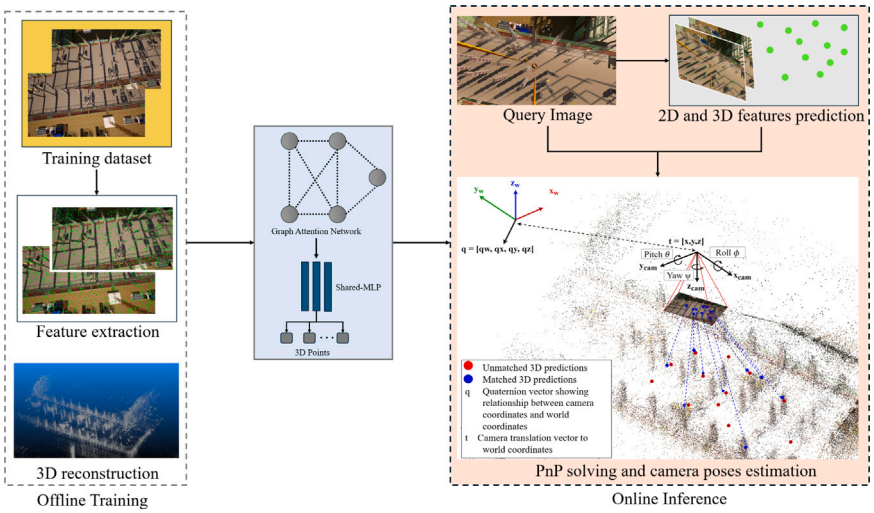


Fig. 2. Visualization of camera re-localization using D2S.

of the construction site are captured both from a camera mounted on the trolley of a tower crane during operation and from a drone-mounted camera during surveying activities. The images, with a resolution of 1920×1080, are suitable for extracting detailed features of large-scale construction projects, while intrinsic parameters may vary depending on the specific camera used on site. To construct a reference scene for supervised learning, we leverage the Structure-from-Motion (SfM) pipeline in COLMAP and carefully filter out frames affected by excessive motion blur or dynamic obstructions. Keyframes are selected based on visual overlap, feature density, and motion stability to ensure accurate 3D reconstruction. The resulting SfM model provides 3D point clouds and camera poses, which are used to extract ground-truth 2D–3D keypoint correspondences for D2S training.

3.1.2. D2S training

Extrinsic parameters play a crucial role because they provide information about the camera position (x_{cam}, y_{cam}, z_{cam}) and orientation (θ, ϕ, Ψ) of relative to the scene, enabling the estimation of the spatial layout of 3D entities. In practice, the orientation is represented by quaternions, which are later converted to rotation matrices for coordinate transformation, as described in Section 3.2.3. To determine 6-DoF camera poses from crane-mounted images, we adopt the D2S framework [42], a lightweight regression-based relocalization method that directly maps sparse 2D keypoints to their corresponding 3D scene coordinates. Instead of relying on dense 3D point clouds and descriptor matching as in traditional feature-matching pipelines, D2S is trained to predict 3D coordinates from keypoint descriptors using a compact neural network. The network architecture consists of a multi-layer graph attention module that allows descriptors to communicate and model spatial context, followed by a shared multi-layer perceptron (MLP) that regresses both 3D coordinates and reliability scores for each keypoint. Each descriptor predicts exactly one 3D point, and the reliability score is trained using a supervised loss function to distinguish robust and stable features from unreliable ones, such as dynamic objects or blurred regions. Ground-truth 2D–3D correspondences are obtained from a Structure-from-Motion (SfM) model built on the training data, and losses are computed by comparing predicted 3D points with the corresponding SfM coordinates, as shown in Fig. 2. Images and SfM point clouds are used to train the D2S model offline. During inference, the pretrained model extracts 2D image features and regresses their corresponding 3D scene coordinates. The predicted correspondences are then solved using a PnP-based algorithm to estimate the camera position $\mathbf{t} = [x, y, z]$ and orientation represented by the quaternion $\mathbf{q} = [q_w, q_x, q_y, q_z]$.

The trained D2S model is then used to estimate new camera poses from unseen query images. Given an input image, keypoints are extracted using SuperPoint, and their descriptors are passed through the trained D2S network to predict their 3D locations and confidence scores. A subset of high-confidence 2D–3D correspondences is selected based on the predicted reliability, and the camera pose is estimated using a RANSAC-based Perspective-n-Point (PnP) algorithm. This inference process avoids the need to store or access dense 3D models, enabling fast and scalable pose estimation. The lightweight nature of the model makes it highly suitable for deployment in tower crane applications, where fast and accurate localization is essential for safe operation.

In the proposed framework, scene coordinate regression is not only used for camera pose estimation, but also serves as the geometric anchor of the overall pipeline. The scene coordinates predicted by the relocalization module define a sparse metric reference space that is later exploited to resolve the scale ambiguity of monocular depth estimation. This role is critical in crane-mounted monitoring, where camera viewpoints vary continuously and explicit recalibration is impractical.

3.2. 3D object localization

This section outlines the processing steps to recover metrically consistent 3D object positions from crane-mounted monocular images, including depth prediction, object detection, and geometric back-projection into the world coordinate system.

3.2.1. Monocular depth map estimation

To estimate the 3D position of detected objects, it is necessary to obtain depth information that relates image observations to the camera coordinate system. Unlike static camera systems where stereo or multi-view geometry can be exploited, crane-mounted monitoring operates under continuously changing viewpoints. Therefore, the system requires a mechanism to infer per-frame depth information from a single image. The Depth Anything model [43] is adopted as the monocular depth estimation backbone to generate dense relative depth predictions for each input image. However, monocular depth estimation inherently suffers from scale ambiguity, meaning the predicted depth values are relative rather than metric. While the model was not explicitly trained on crane-mounted datasets, its training set includes drone and elevated-view images, making it suitable for similar top-down perspectives. As a result, this model allows the system to recover generalized per-pixel depth information without the need for stereo cameras or LiDAR sensors, making it well-suited for crane-mounted monocular setups.

Algorithm 1 Sparse-Anchor depth scale alignment.

Require: Dense relative depth map D_{rel}, sparse metric depth anchors D_{sparse}, RGB image I

Ensure: Metric-consistent dense depth map D_{final}, global scale factor s

1: Identify valid overlapping pixels Ω between D_{rel} and D_{sparse}
2: Compute scale ratios $r_i = D_{sparse}(i)/D_{rel}(i)$ for all $i \in \Omega$
3: Estimate the global scale factor s using the median of $\{r_i\}$
4: Align the dense depth map to metric scale: $D_{scale} = s \cdot D_{rel}$
5: Initialize fused depth map $D_{metric} \leftarrow D_{scale}$
6: Replace values in D_{metric} at $i \in \Omega$ with $D_{sparse}(i)$
7: $D_{final} \leftarrow \text{GuidedFilter}(D_{metric}, I)$

A key challenge of monocular depth estimation is the inherent ambiguity of absolute scale. As a result, the predicted depth map provides only relative depth values and cannot be directly used for metric 3D localization. This limitation becomes particularly critical in crane-mounted monitoring scenarios, where accurate spatial relationships between workers and suspended loads must be computed in metric space. The scale ambiguity problem in monocular depth estimation has been extensively studied in prior work. Early approaches such as Eigen et al. [44] learn depth relationships using scale-invariant training objectives, while later methods estimate depth jointly with camera motion through self-supervised learning frameworks [45,46]. These approaches have demonstrated that reliable dense depth structures can be recovered from monocular imagery. However, the predicted depth values are typically defined only up to an unknown global scale, which limits their direct use for metric 3D localization in real-world environments.

To recover metric scale required for spatial reasoning, the proposed framework augments the monocular depth prediction with geometric constraints derived from the reconstructed scene. Specifically, sparse 3D points obtained from the SfM reconstruction are reprojected into the image plane using the estimated camera pose. This reprojection produces sparse but metrically accurate depth anchors at selected pixel locations. These anchors serve as reliable geometric references that allow the dense monocular depth map to be aligned with the metric scene coordinate system, enabling absolute-scale depth recovery while preserving the dense spatial structure predicted by the monocular depth estimator.

Let D_{rel} denote the dense relative depth predicted by the monocular depth estimator, and D_{sparse} denote the sparse metric depth anchors obtained through reprojection of 3D scene points.

We first define the set of valid overlapping pixels.

$$\Omega = \{i \mid D_{sparse}(i) > 0 \wedge D_{rel}(i) > 0\} \tag{1}$$

which represent reliable geometric correspondences between the sparse metric depth and dense relative depth predictions.

A global scale factor s is then estimated as

$$s = \text{median}_{i \in \Omega} \left(\frac{D_{sparse}(i)}{D_{rel}(i)} \right) \tag{2}$$

The dense depth map is subsequently aligned to the metric scale

$$D_{scale} = s \cdot D_{rel} \tag{3}$$

Finally, sparse metric anchors are fused with the scaled dense depth map to preserve geometric consistency while maintaining dense spatial coverage.

This reprojection-based scaling strategy enables the dense monocular depth predictions to be transformed into metric space while maintaining geometric consistency with the reconstructed scene coordinates. The complete reprojection-based depth scale alignment procedure is summarized in Algorithm 1.

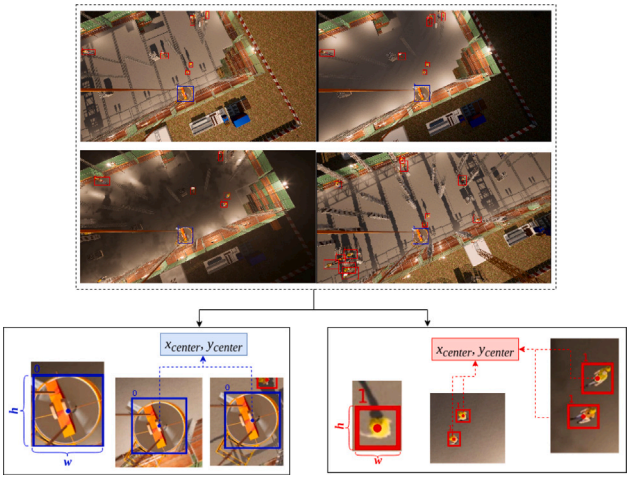


Fig. 3. Results of 2D detection from RTMDet.

3.2.2. 2D object detection with top-down view camera

Object detection is a fundamental step in 3D estimation, as it provides object classes, identities, and pixel-level locations, which are subsequently transformed into 3D coordinates. Detecting workers or construction machinery from a crane-mounted camera presents unique challenges due to the elevated top-down viewpoint, scale distortion, and frequent occlusions. While most object detection models are trained on horizontal or vehicle-mounted perspectives, recent studies have shown that modern detectors based on RetinaNet [36,47] and YOLO [12,48] can perform effectively on aerial and top-down views captured by crane-mounted cameras or drones . In this study, we adopt a one-stage detector, Real-Time Model of Object Detection (RTMDet) [49], to achieve a balance between detection accuracy and inference efficiency. Compared with existing one-stage detectors, RTMDet leverages large-kernel convolutions and a more balanced computational design between the backbone and neck, enabling efficient contextual feature extraction while maintaining high inference speed without sacrificing accuracy. Therefore, it is well suited for detecting small and occluded objects in top-down construction scenarios.

The RTMDet model is trained on a large-scale dataset constructed from images collected from crane-mounted cameras and drones, along with publicly available surveillance data from the VisDrone dataset [50]. To further enhance data diversity, a synthetic dataset representing realistic construction environments is generated under varying weather and lighting conditions, simulating different object scales and noise patterns observed on actual construction sites. The model is then trained on a hybrid dataset comprising both real and synthetic images.

After training, the model outputs bounding boxes in the form of (x_{center}, y_{center}, w, h) in pixel coordinates. Given the top-down viewpoint, the positions of workers and lifted objects are approximated using the center points of the bounding boxes (x_{center}, y_{center}), as illustrated in Fig. 3. These points are used as representative image coordinates *(u, v)* for subsequent back-projection, as described in Section 3.2.3.

3.2.3. 3D localization

Building upon the detected 2D object locations, most existing methods [8,51,52] estimate 3D worker locations (x_w, y_w, z_w) by simplifying the computation and assuming a fixed object elevation z_w in world coordinates, followed by projecting 2D observations into 3D space. In this study, we propose a method inspired by Han et al. [12] that enables direct estimation of all three coordinates (x, y, z) without requiring any assumption about object elevation. Our method incorporates a depth

Table 2
Risk assessment matrix.

Status	Explanation	Risk assessment level
$d \leq r_{crit}$	Worker is located within the fall zone and must immediately vacate the area	**Warning**
$r_{crit} < d \leq r_{med}$	Worker is in close proximity to the fall zone and may enter a hazardous region	**Caution**
$d > r_{med}$	Worker is positioned in a safe zone with minimal risk exposure	**Safety**

map for each frame as an additional cue, where each pixel value represent the distance to the corresponding object. Subsequently, monocular depth estimation, camera calibration, and geometric reprojection are combined to recover full 3D positions of detected objects in the world coordinate system. In addition, when images are captured from a large distance, lens distortion factors k_1 and k_2 should be considered, as they may bend and stretch objects near image boundaries, leading to inaccurate localization in subsequent steps. To mitigate this effect, a Region of Interest (ROI) is defined around the image center to exclude distorted detections. According to photogrammetric and optical standards [53,54], distortion is considered non-negligible when the relative distortion threshold ε exceeds 0.5%–1% of the image height. Therefore, in this study, a conservative threshold of 0.3% is adopted, and the ROI dimensions are defined as follows:

$$W_{ROI} = \frac{2\sqrt{\dfrac{-k_1 + \sqrt{k_1^2 + 4k_2\varepsilon}}{2k_2}}}{\sqrt{\frac{1}{f_x^2} + \frac{1}{\alpha^2 f_y^2}}}, \qquad H_{ROI} = \frac{W_{ROI}}{\alpha}. \tag{4}$$

where W_{ROI}, H_{ROI} are the width and height of ROI, and α is the aspect ratio.

The camera intrinsic matrix is first modeled to define the relationship between pixels in the image plane and camera-centered coordinates. The intrinsic $\mathbf{K} \in \mathbb{R}^{3\times3}$ matrix is determined from camera specifications, including focal lengths (f_x, f_y) and principal point offsets (c_x, c_y) and then is expressed as:

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{5}$$

For each target object detected in Section 3.2.2, the corresponding depth value is estimated as a median depth of pixels within each bounding box. Once the depth value is obtained, we use the camera intrinsic matrix $\mathbf{K}$ to back-project representative pixel points $\mathbf{p} = [u, v, 1]^\top$ to normalized camera coordinates, and apply the depth $\mathbf{w}$ to scale the point along the camera ray to its actual 3D location:

$$\mathbf{X}_{cam} = \mathbf{w} \cdot \mathbf{K}^{-1}\mathbf{p} \tag{6}$$

Because the recovered depth values are aligned with the metric scene coordinate space, the back-projection process yields metrically consistent 3D object locations in the world coordinate system. This property is essential for crane-mounted monitoring, where dynamic viewpoint changes make repeated calibration impractical.

To convert the points into the global world system, both intrinsic and extrinsic parameters are applied. While the intrinsic matrix represents the internal properties of an image and its mapping from image pixels to the local camera coordinate system, the extrinsic matrix provides spatial information about the camera coordinate system in the world reference frame. These parameters are essential for accurately converting object positions from local camera coordinates to global world coordinates. From the D2S relocalization model, we extract the rotation and translation of each frame. The rotation is represented as a quaternion $q = [q_w, q_x, q_y, q_z]$, as illustrated in Fig. 2, while $t = [x, y, z]$ describes the position of the camera in world coordinates. Based on these values, the extrinsic components of the frame are computed as follows:

$$\mathbf{R} = \begin{bmatrix} 1 - 2(q_y^2 + q_z^2) & 2(q_x q_y - q_w q_z) & 2(q_x q_z + q_w q_y) \\ 2(q_x q_y + q_w q_z) & 1 - 2(q_x^2 + q_z^2) & 2(q_y q_z + q_w q_x) \\ 2(q_y q_z + q_w q_y) & 2(q_y q_z + q_w q_x) & 2(q_x^2 + q_y^2) \end{bmatrix} \in \mathbb{R}^{3\times3} \tag{7}$$

$$\mathbf{t} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \in \mathbb{R}^{3\times1} \tag{8}$$

With the extrinsic transformation obtained from the D2S module and the defined intrinsic matrix, the formula to transform a 3D point from the local camera system to the global world space is:

$$\mathbf{X}_{cam} = \mathbf{R}\mathbf{X}_{world} + \mathbf{t} \tag{9}$$

This transformation moves the 3D point from the camera coordinate system into the global world frame by first translating it relative to the camera's position and then rotating it into the world orientation using the inverse rotation matrix. The process is repeated for each detection, allowing us to recover complete 3D positions *(x, y, z)* of all objects in a unified world coordinate system.

3.3. Digital twin visualization

After recovering metrically consistent 3D object positions, the proposed framework establishes a digital-twin-based safety monitoring pipeline for spatial registration, hazard-zone estimation, and rule-based risk assessment.

3.3.1. Spatial registration

Although 3D worker coordinates can be recovered from camera re-localization and depth maps, these positions are originally defined in the local reference frame of the SfM-based D2S space. In contrast, the BIM model imported via the Revit–Omniverse connector preserves its own global metric coordinate system with real-world orientation, unit scale, and origin definition. Therefore, a synchronization step is introduced to establish a transformation between the SfM coordinate system and the BIM coordinate system. Through this transformation, both the estimated camera poses and the reconstructed 3D worker coordinates can be consistently represented in the BIM-based real-world space, enabling subsequent safety monitoring and validation as described in Section 3.3.4 and Section 4.

The registration problem can be formulated as estimating a rigid transformation matrix $\mathbf{T} \in \mathbb{R}^{4\times4}$ that minimizes the discrepancy between the point cloud generated by D2S and the BIM geometry. To achieve this, salient geometric features are extracted from both datasets. Candidate correspondences are then established using feature descriptors and matched via nearest-neighbor search. Since these matches may contain outliers, a RANSAC-based estimator is first used to compute a robust initial alignment. Finally, the transformation is refined using the Iterative Closest Point (ICP) algorithm, which iteratively minimizes point-to-plane distances until convergence. The resulting matrix $\mathbf{T}$ is subsequently applied to transform all relocalized worker positions and safety zones into the BIM coordinate system, enabling spatial synchronization within the digital twin. The relationship between the transformation matrix, the reconstructed point cloud space, and the BIM coordinate space is illustrated in Fig. 4.

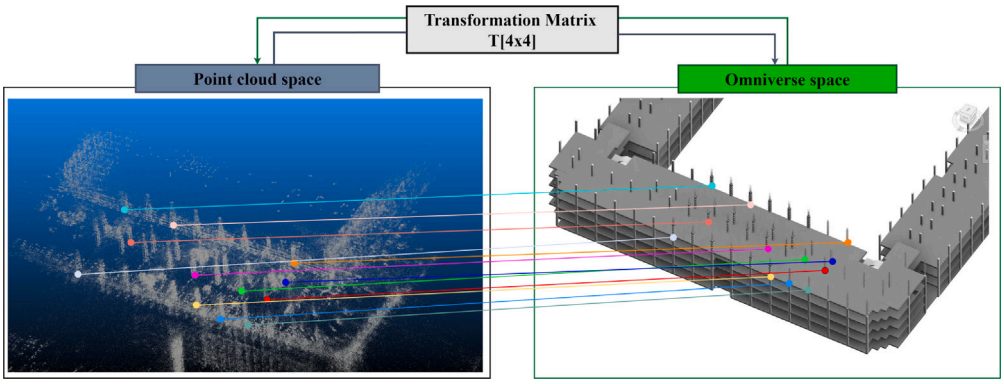


Fig. 4. Spatial registration between point cloud and BIM model using a transformation matrix.

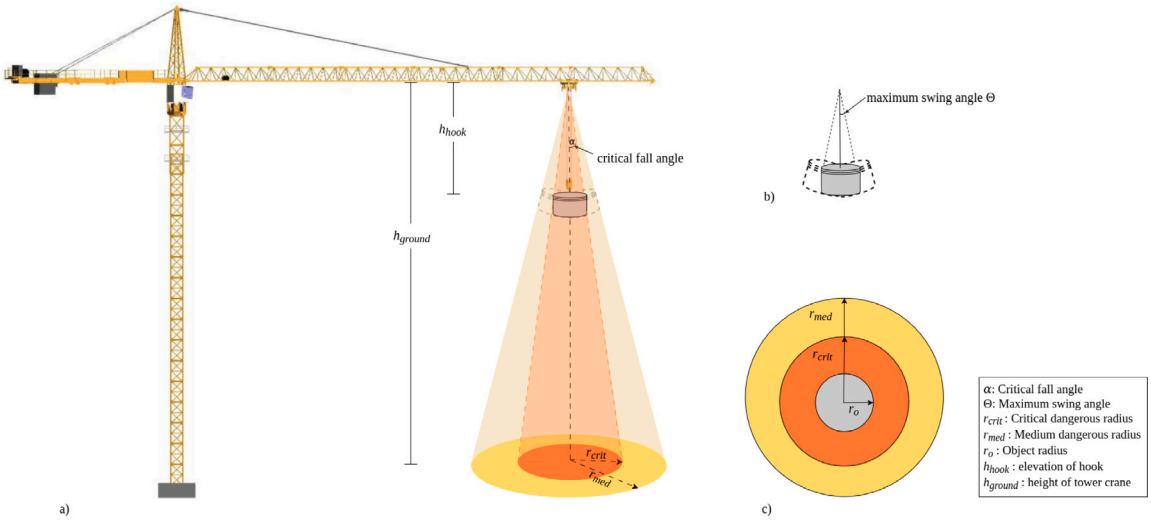


Fig. 5. Illustration of dangerous fall zone of tower crane (a), maximum swing angle of object oscillation (b) and projection of dangerous areas on ground (c).

3.3.2. Crane safety area estimation

In the following step, overhead regions with high and medium risk due to falling objects during tower crane operation are determined. These regions are modeled as conical zones, whose projections on the ground define controlled areas to prevent accidents. Based on the dimensions and lifting elevation of the object, the critical dangerous radius can be calculated as follows:

$$\frac{r_o}{r_{crit}} = \frac{h_{hook}}{h_{ground}} \tag{10}$$

Here, r_o is the maximum geometric radius of the object, h_{hook} is the metric depth value extracted from 2D detection in Fig. 5, while h_{ground} is the z_w obtained from the camera relocalization process. The critical dangerous radius varies dynamically with respect to the elevation; that is, the hazard zone expands when the object is lifted higher and decreases as the object approaches the ground.

In addition, the medium-risk zone is estimated based on the oscillation amplitude during operation. In practice, this factor is influenced by various conditions in real-world environments. Therefore, in this study, a fixed value is adopted for the maximum allowable swing angle $\Theta = 2.5°$, as suggested by Dutta et al. [55], to simplify the calculation. Accordingly, the medium-risk area is defined as follows:

$$r_{med} = h_{ground} \cdot \tan(\alpha + \Theta) \tag{11}$$

where

- $\alpha = \tan^{-1}\left(\frac{r_o}{h_{\text{hook}}}\right)$ is the critical fall angle
- $\Theta = 2.5°$ is the maximum allowable swing angle due to oscillation

Finally, the safety zones are not only revealed in 3D space but also are back-projected onto each image frame based on the Eq. (9).

3.3.3. Omniverse-based deployment

After achieving synchronization, the 3D localization data are directly integrated into NVIDIA Omniverse to enable Digital Twin (DT) functionality. NVIDIA Omniverse is a state-of-the-art development platform for building and operating 3D applications and services at scale 1u [56], and has been proven to be a potential solution for digital integration among various types of data [41]. In this study, we adopt NVIDIA Omniverse USD Composer [57] as the runtime environment for interactive site visualization and safety assessment. USD Composer provides advanced scene assembly, high-fidelity RTX rendering, and Python scripting support, enabling seamless mapping of camera poses and depth-derived 3D worker positions into the digital twin environment. This capability not only facilitates accurate spatial synchronization but also supports dynamic simulation of worker movements and potential collision scenarios.

In NVIDIA USD Composer (version 2023.2.5) [57], the prebuilt 3D BIM model is directly imported while its original coordinate reference and scale are preserved through the Omniverse Revit Connector. This ensures a one-to-one spatial correspondence between the real-world construction site and the digital twin environment. Subsequently, all 3D entities, including worker positions and projections of hazard fall zones, are aligned from the D2S relocalization space to the Omniverse USD Composer environment by applying the transformation matrix **T**. Simultaneously, the Replicator API, supported by the Omniverse

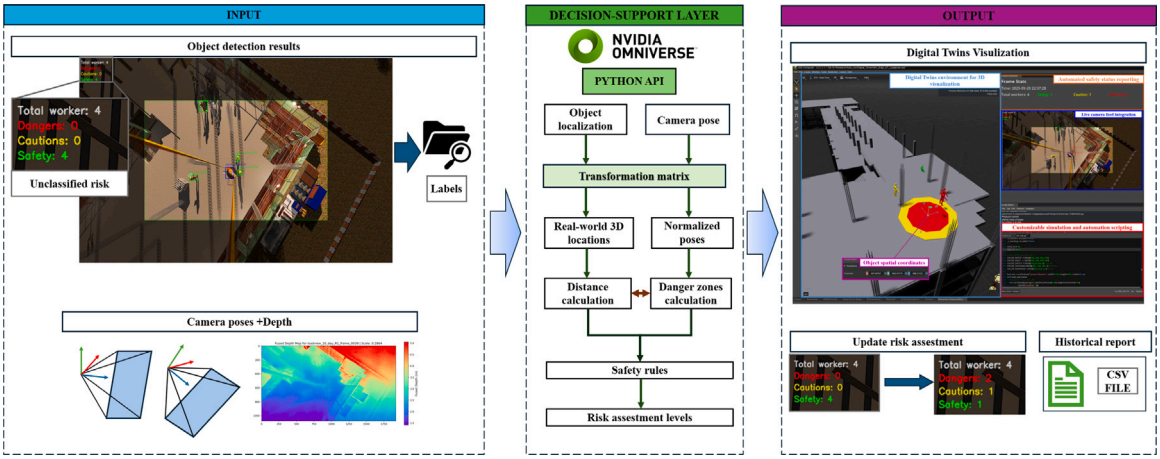


Fig. 6. Digital Twin-based decision-support pipeline for rule-based safety risk assessment and visualization.

extension package, is used to read and record object attributes such as class, translation, and risk level, whose calculation process is described in Section 3.3.4. In addition, a visualization window displaying live video with 2D detections is implemented, enabling a double-check procedure between recorded data and visualization results.

3.3.4. Risk assessment

In the final stage of digital twin illustration, risk level assessment is performed to quantify safety conditions around the crane operation zone. Beyond visualization, the digital twin performs rule-based spatial reasoning and risk evaluation on vision-derived 3D data. Instead of computing safety distances externally and re-importing the results, a Decision-Support Layer is established via the Python API in USD Composer to compute Euclidean distances and perform rule-based risk evaluation directly within the digital twin environment, thereby reducing latency and data transfer overhead. These distances are then compared with the critical dangerous radius r_{crit} and medium dangerous radius r_{med} to define the risk level for each entity, as detailed in Table 2. In addition, each worker is annotated with a specific color based on the defined risk level using the Material Definition Language (MDL) extension in NVIDIA Omniverse USD Composer, while a Python script is used to summarize all hazard classification results and report the current safety status. In parallel, the risk assessment is also updated in the 2D detection camera window, ensuring consistent visualization in both 2D and 3D. In the final step, a historical report is automatically generated; this file records all worker information in hazard zones, including date, time, positions, and safety status for further analysis. The process is repeated whenever data from a new frame are updated, ensuring real-time monitoring capability, as illustrated in Fig. 6.

4. Experiments

In this section, we conduct multiple experiments to evaluate the performance of camera relocalization in the D2S-MDE framework and 2D object detection using RTMDet under top-down views with small and overlapping entities on construction sites. Finally, the proposed framework is applied in a case study of crane safety operations, where workers are detected from a dynamic trolley-mounted camera, localized in 3D space, and integrated into a digital twin for combined detection, localization, and automated risk assessment reporting. All procedures are performed on a computer equipped with an NVIDIA RTX 4090 GPU, an Intel i9-13900KF processor, and 64 GB of RAM.

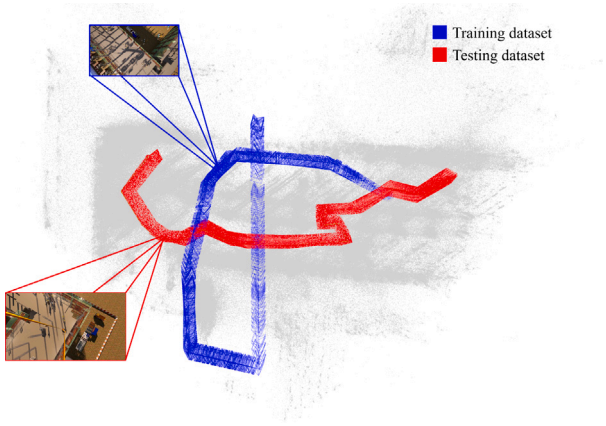


Fig. 7. Visualization of training and testing camera path.

4.1. Scene coordinates regression model performance

In order evaluate the performance of camera relocalization, we conduct experiment by training multiple datasets including both simulation and a real construction project using D2S-based platform [42], as listed in Fig. 8. In the first experiment, we set up a testbed at a construction site located in Miryang-si, Gyeongsangnam-do, Republic of Korea. Aerial-view videos of the scene were captured using a DJI Mini 2 Pro drone with a resolution of 1920 × 1080, and the dataset was divided into 75% training data and 25% testing data. In the second experiment, we evaluate model performance on synthetic datasets representing construction activities generated by Twinmotion. This software can replicate realistic structural elements and visual variations under different weather conditions, making it suitable as a testbed for vision-based deep learning models. In this study, we simulate a camera trajectory that mimics drone-based surveying to generate synthetic datasets under daytime conditions with clear skies, ensuring that D2S can capture as many scene features as possible (Fig. 7). After that, the testbed includes videos recorded from trolley-mounted cameras in varied time milestones and three basic weather conditions. This experimental setup reflects real construction scenarios, where site surveying activities are typically conducted once or twice per week, while heavy machinery operations occur more frequently. The principle of the second experiment is that the datasets used for training and testing in both real and synthetic scenarios are recorded along independent trajectories, where the training set is used for scene reconstruction and the testing set is applied for safety-related evaluation. Therefore, the model performance must generalize across varying conditions on construction sites.

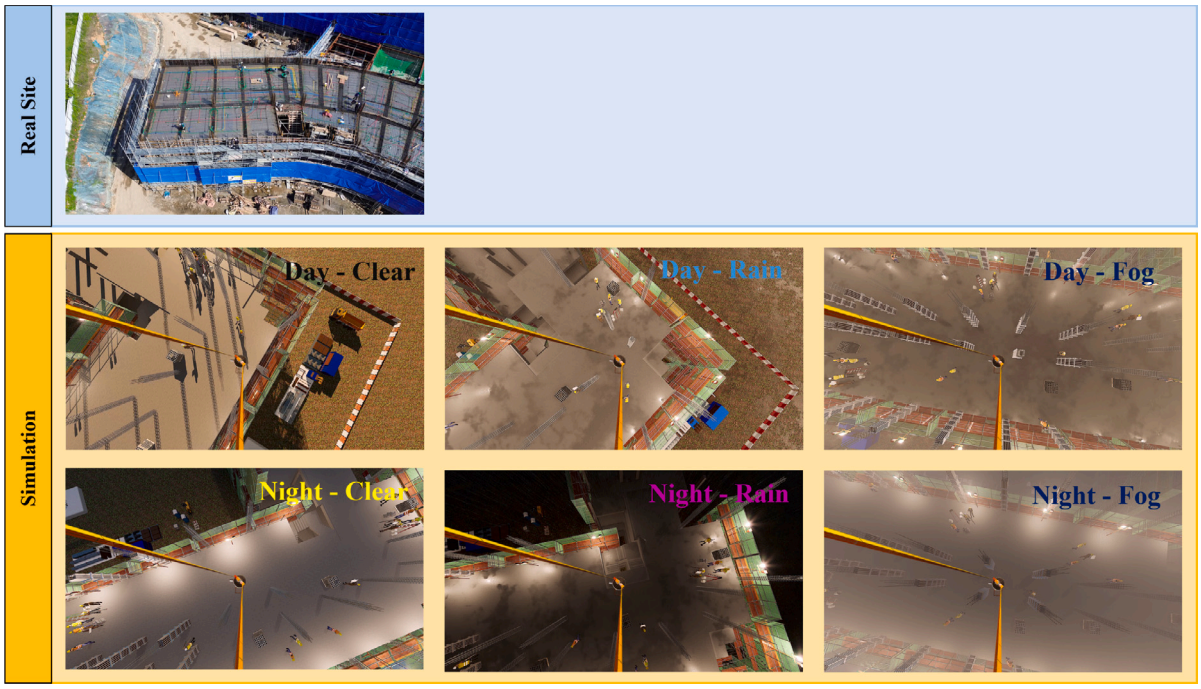


Fig. 8. Visualization of different datasets for D2S relocalizer evaluation.

Furthermore, in this study, reference camera poses are obtained from the same SfM reconstruction used to generate sparse 3D point cloud labels for the D2S training process. The camera trajectories estimated by the COLMAP SfM pipeline are treated as pseudo ground truth and serve as reference poses for evaluating the camera relocalization performance. COLMAP has been shown to achieve state-of-the-art performance in large-scale scene reconstruction [58]. Specifically, during SfM reconstruction, all 4500 input images were successfully registered, producing a sparse point cloud with 72,890 3D points. The reconstruction achieved a mean track length of 117.99 and a mean reprojection error of 0.81 pixels (median 0.61 pixels), indicating stable image registration and strong geometric consistency of the reconstructed camera poses and 3D structure. These statistics suggest that the reconstructed camera poses are sufficiently stable to serve as reference poses for evaluating the relocalization accuracy.

4.1.1. Qualitative results

Figs. 9, 10, and 11 illustrate the qualitative performance of the D2S-MDE framework under both real and synthetic environments. As shown in Fig. 9, the camera relocalization results on a real construction site demonstrate that the predicted camera poses (red) are almost identical to the ground-truth trajectory (green). Despite the inherent complexity of construction environments, including dynamic workers and equipment, the framework is able to reconstruct a continuous camera path with minimal drift. This confirms that the proposed D2S-MDE framework is well suited for clear-visibility conditions within the same domain, which correspond to the primary operational settings of crane-mounted and aerial cameras on construction sites. Fig. 10 illustrates performance under moderately degraded conditions, including day-rain, day-fog, and night-clear scenarios. In the rain scenario, most predicted camera poses remain close to the ground truth in both position and orientation, indicating moderate robustness to visual noise caused by rain streaks and surface reflections. Under foggy daytime conditions, larger pose deviations are observed, likely due to reduced visibility of structural features, which weakens the reliability of scene coordinate regression. Nevertheless, for a subset of frames, the overall trajectory structure is still preserved, suggesting partial robustness when degradation is not severe. Similarly, in night-clear scenes, the system is able to recover camera poses for the majority of frames, confirming that sufficient illumination alone can still support reliable localization.

Fig. 11 highlights the most challenging conditions, characterized by both low illumination and severely reduced visibility. In night-fog scenarios, localization performance degrades substantially, with frequent pose failures and large deviations from the reference trajectory. This behavior indicates that the proposed vision-based relocalization pipeline reaches its operational limits when critical visual constraints are lost. These results emphasize that, while the framework performs reliably under clear or moderately degraded conditions, its applicability under extreme weather and lighting conditions remains limited.

Overall, these qualitative results highlight a clear relationship between training data characteristics and deployment performance. As described in the data preparation stage, the D2S relocalizer is primarily trained using drone-captured and synthetic overhead videos recorded under clear daytime conditions, which provide rich structural features and stable illumination. Consequently, the framework achieves its highest accuracy and trajectory consistency when tested on scenes that share similar visual conditions, such as real construction sites and synthetic day-clear scenarios. When applied to out-of-domain environments, including different weather or illumination conditions, the pipeline can still function under certain constraints. In particular, night-clear scenes remain partially compatible with the learned scene representations, as sufficient structural visibility is preserved despite the absence of daylight. However, scenes involving severe visibility degradation, such as dense fog or combined night-fog conditions, lead to substantial performance degradation due to the loss of reliable visual cues required for scene coordinate regression. These observations indicate that the proposed pipeline inherently favors clear-sky conditions and performs most reliably when the training and testing domains are visually consistent. While limited generalization to unseen domains is possible under clear visibility, the results confirm that clear-weather scenes represent the preferred operational regime of the current D2S-MDE framework.

4.1.2. Quantitative results

Table 3 presents a comprehensive quantitative evaluation of camera relocalization performance across different environmental conditions.

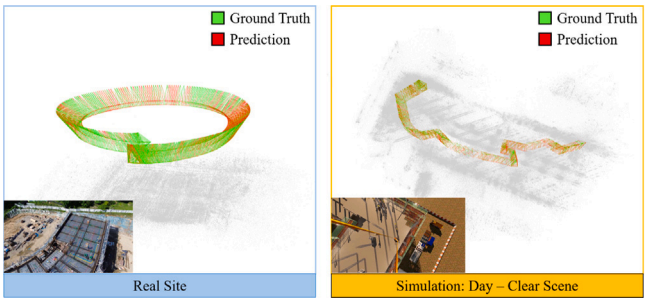


Fig. 9. Visualization of camera relocalization results from real construction site and simulation day-clear scenes.

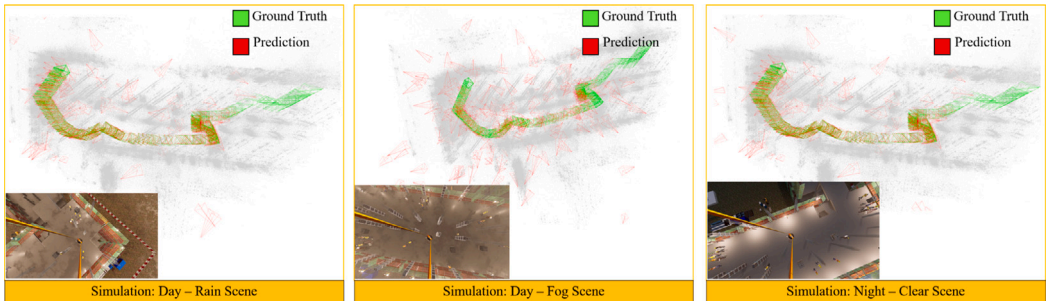


Fig. 10. Visualization of evaluation results from simulation day-rain, day-fog and night-clear scenes respectively.

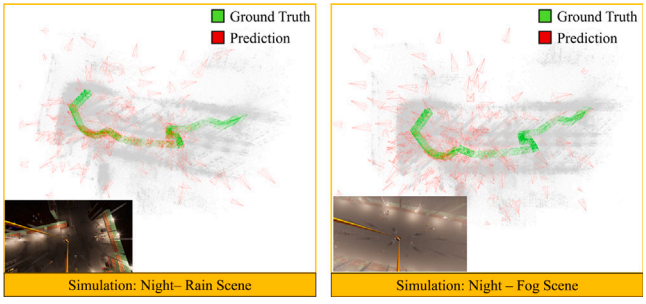
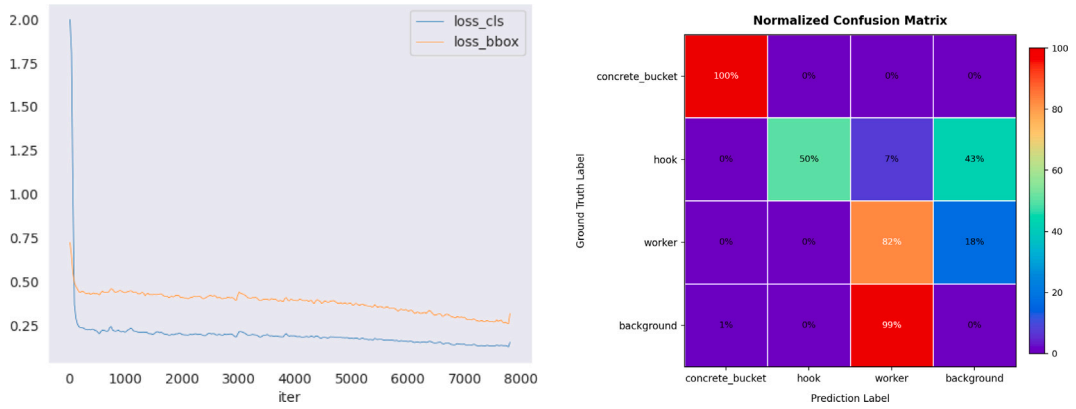


Fig. 11. Visualization of evaluation results from simulation night-rain and night-fog scenes.

To characterize both typical accuracy and robustness, we report the median (P50) and P75 translational and rotational errors, together with explicit success and failure rates. The median (P50) reflects the typical localization accuracy achieved under a given condition, while the P75 error captures a robustness margin by indicating the error level below which the majority of successfully localized frames fall. This choice avoids overemphasizing extreme tail values that are often dominated by degenerate PnP solutions, and instead provides a stable and interpretable summary of localization performance for safety-critical deployment. A frame is considered successfully localized if a valid camera pose can be recovered via PnP using the predicted scene coordinates with at least 20 inlier correspondences. Otherwise, the frame is classified as a failure and excluded from error statistics. Higher error percentiles are therefore not explicitly reported, as such cases are more appropriately reflected by the reported failure rates rather than by large numerical error values that do not correspond to usable camera poses. On the real construction site, the proposed framework achieves a median translation error of 0.012 m and a median rotation error of 0.019°, with a P75 translation error of 0.017 m and a 100% localization success rate. These results confirm that the system can reliably recover camera poses from real-world data captured under practical deployment conditions. Similarly, under synthetic day–clear conditions, performance remains highly consistent, with errors of 0.016 m/0.030° at P50, indicating strong generalization between real and simulated domains when visual conditions are favorable.

As environmental conditions degrade, localization accuracy naturally decreases. In the day-rain scenario, the median translation error increases to 0.379 m with a rotation error of 0.858°, while day-fog conditions present further degraded localization performance, with 0.973 m/2.396° at P50. Nevertheless, the P75 errors remain within a controlled range, suggesting that a substantial portion of successfully localized frames still preserve usable geometric accuracy. Nighttime sequences exhibit a similar trend, with moderate deviations of 0.321 m/0.724° under night-clear conditions. Meanwhile, adverse weather leads to more pronounced degradation, particularly in night-rain and night-fog scenarios. In these cases, failure rates increase significantly, reflecting insufficient visual constraints for reliable PnP estimation rather than silently producing erroneous poses.

The observed behavior is consistent with the original D2S framework [42], which demonstrated robustness to viewpoint changes by training and testing on disjoint trajectories in datasets such as 7-Scenes, Cambridge Landmarks, and BKA. This property is particularly relevant to our setting, where training data are collected from top-down crane-mounted cameras or drone-like aerial viewpoints during daytime surveys, while testing may occur along different paths and under distinct environmental conditions. The strong real-site performance, together with controlled degradation under out-of-distribution scenarios, indicates that the proposed system inherits D2S's ability to generalize across trajectories and viewpoints. However, these error values also reinforce the limitations discussed in Section 4.1.1. While the



(a) Training loss curves of RTMDet (b) Normalized confusion matrix

Fig. 12. Training and evaluation results of the RTMDet model on the custom dataset.

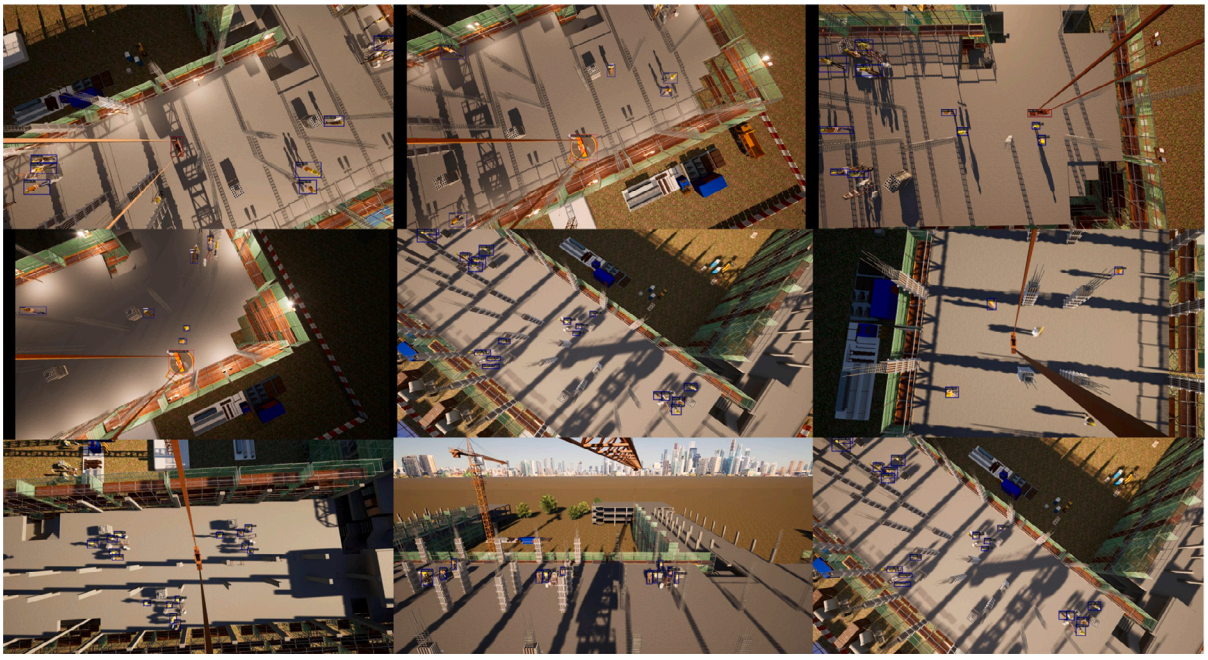


Fig. 13. Samples of qualitative detection results from RTMDet.

framework achieves strong performance under favorable conditions, degradation in camera pose predictions in more challenging scenarios can negatively affect the depth scaling procedure and subsequent 3D localization.

Overall, these results show that the scene coordinate regression-based relocalization model can achieve high accuracy for both crane CCTV and aerial drone perspectives, validating its applicability under clear-weather conditions. At the same time, the explicit reporting of P50–P75 error ranges and failure rates highlights the system's sensitivity to severe weather and lighting changes. The rapid increase in error magnitude and failure rates under foggy and nighttime scenarios quantitatively defines the operational limits of the proposed approach. These findings support a clear distinction between the system's validated use under favorable visual conditions and its reduced reliability under severe visibility degradation.

4.2. Worker detection

Regarding object detection results, the RTMDet-L model was trained using the custom dataset described in Section 3.2.2 and tested on crane-mounted top-down images, including three classes: worker, hook, and concrete bucket. The training was conducted over 300 epochs using CSPNeXt (P5) with large-kernel convolutions, the AdamW optimizer with a learning rate of 0.005 and weight decay of 0.05, along with combined augmentation strategies.

Fig. 12(a) presents the training loss curves, showing a steady decrease in both classification and bounding box regression losses, which indicates stable convergence during training. In addition, the confusion matrix in Fig. 12(b) demonstrates the accuracy of the detector in distinguishing between object classes. The results show that workers and concrete buckets are detected with high accuracy, while the hook class remains more challenging due to its small size and frequent

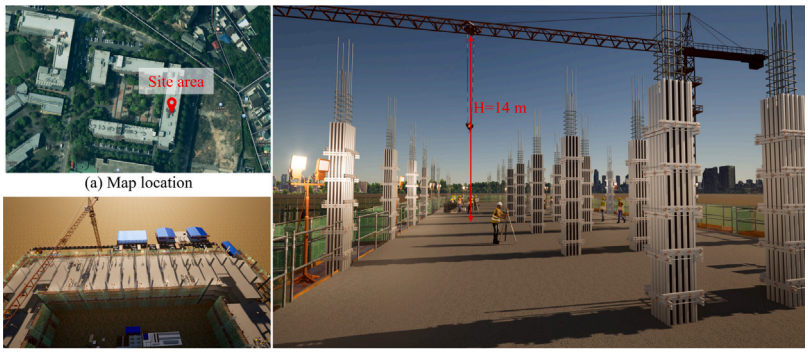


Fig. 14. Overview of case study for pipeline testbed.

Table 3
Camera relocalization accuracy, error distribution (P50–P75), and failure rates across different environmental conditions.

Metric	Real-site	Day-clear	Day-rain	Day-fog	Night-clear	Night-rain	Night-fog
Median Error (m)	0.012	0.016	0.379	0.973	0.321	0.724	2.809
Median Error (deg)	0.019	0.030	0.858	2.396	0.724	1.384	7.362
P75 Error (m)	0.017	0.023	0.808	2.103	0.581	1.462	28.700
P75 Error (deg)	0.027	0.044	1.88	5.42	1.37	2.76	66.1
Success Rate (%)	100.0	100.0	76.1	53.4	82.8	38.2	27.3
Failure Rate (%)	0.0	0.0	23.9	46.6	17.2	61.8	72.7

occlusions in cluttered backgrounds. Besides quantitative evaluation, the qualitative results in Fig. 13 show that RTMDet is able to detect all entities across different scenarios. In particular, worker detection remains stable across various scenes, even in small and overlapping cases. In contrast, the hook class is sometimes incorrectly detected, likely due to dataset imbalance, whereas the concrete bucket is accurately recognized. These results provide evidence of the stable performance of RTMDet on top-down view datasets, thereby validating the applicability of the model for construction safety monitoring and as the detection module for subsequent 3D localization and risk assessment in the proposed framework.

4.3. Case study

Finally, we apply the proposed method in a testbed at the Engineering 2 building located at Sungkyunkwan University, Suwon-si, South Korea, as shown in Fig. 14. We reconstruct a construction site in Twinmotion and replicate safety scenarios to monitor worker activities within the tower crane operating range. The use of synthetic environments and simulation-based data generation has been widely explored in previous construction vision studies [59,60] to address limited real-world data availability and improve model robustness under diverse site conditions. Using a camera mounted on the crane trolley at a height of 14 m above the monitored area, workers and the concrete bucket are detected and transformed into 3D positions via the D2S-MDE pipeline described in Section 3. In addition, their positions are linked to NVIDIA Omniverse for safety zone estimation, risk classification, and safety report generation using a digital twin framework. To evaluate the applicability of the D2S-MDE framework and the digital twin approach for on-site safety monitoring, the pipeline is tested using a 90-second video recording of the concrete pouring phase, during which the crane carries a concrete bucket over workers on the construction site. The day-clear scene, which is reported with lowest error and highest success rate, is used to eliminate uncertainties. Our target is to evaluate the effect of scaled depth map on 3D localization.

4.3.1. 3D localization evaluation

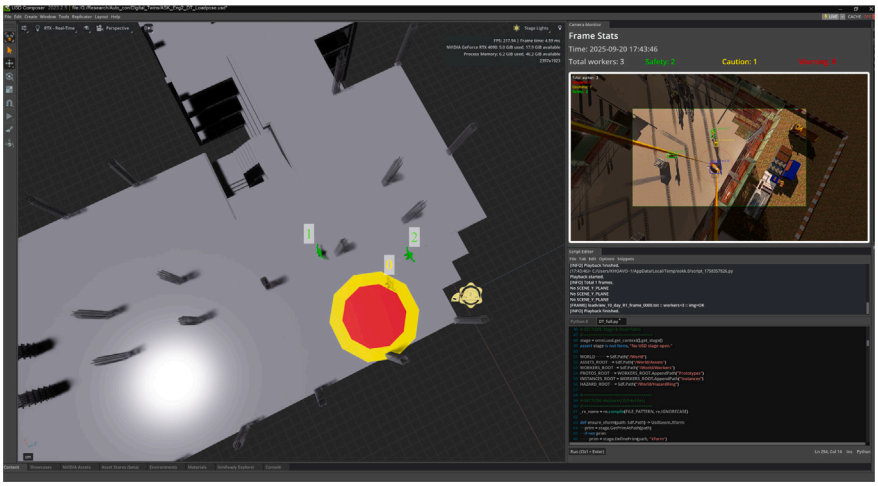
After the estimation step, the predicted 3D positions $(x_{\text{pred}}, y_{\text{pred}}, z_{\text{pred}})$ are evaluated against the ground truth $(x_{\text{gt}}, y_{\text{gt}}, z_{\text{gt}})$ using the Root Mean Square Error (RMSE), which measures the Euclidean distance between predicted and reference positions in meters. To enable fair comparison across different camera heights, we additionally define a Height-Normalized Localization Score (HNA) as a normalized, application-oriented performance indicator rather than a conventional accuracy metric. The HNA reflects relative localization reliability with respect to the camera-to-ground distance. In this study, RMSE serves as the primary quantitative metric for 3D localization performance, while HNA is reported as a supplementary indicator to facilitate intuitive interpretation in crane-mounted safety monitoring scenarios. These two metrics are described as follow:

$$\text{RMSE} = \sqrt{(x_{pre} - x_{gt})^2 + (y_{pre} - y_{gt})^2 + (z_{pre} - z_{gt})^2} \tag{12}$$

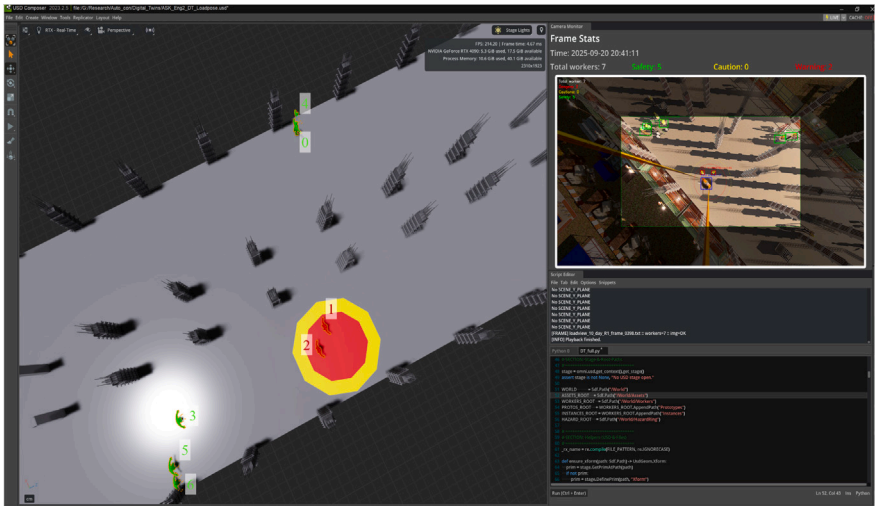
$$\text{HNA} = \max\left(0, \left(1 - \frac{\text{RMSE}}{H}\right)\right) \times 100\%, \tag{13}$$

where H denotes the height from the camera to the building floor. This normalization reflects the intuition that the farther the camera is from the objects, the more sensitive the localization becomes to noise; therefore, the evaluation should be scaled accordingly. Figs. 15(a) and 15(b) illustrate the visualization of 3D localization in the digital twin platform. The results indicate that workers positioned farther from the image center in the 2D domain are localized with reduced accuracy compared to those near the center, while some individuals are partially occluded by structural elements. Although a region of interest (ROI) is applied to alleviate geometric distortion, a small amount of pixel information may still be affected, particularly when images are captured from greater distances. In addition, the use of bounding box centroids as representative points introduces minor discrepancies. Points near the image center are generally well projected onto the ground plane, whereas those toward the periphery may correspond to less reliable depth values. These factors may introduce modest deviations in accuracy. Nonetheless, workers situated near the safety zone remain well localized, allowing the pipeline to perform reliable risk assessment.

Tables 4(a) and 4(b) summarize the errors in terms of RMSE (m) and the derived Height-Normalized Localization Score (HNA) of corresponding worker IDs in Fig. 15(a) and Fig. 15(b), respectively. The ground-truth extracted from Twinmotion is used to compare with the predicted localization results, distance from current location to hazard center as well as the distance-based risk level. From these tables, the



(a) Frame 0



(b) Frame 398

Fig. 15. Samples of qualitative results for 3D localization.

Table 4
Comparison between predicted and ground-truth positions with risk assessment. Distances are in meters.

(a) Frame 0 with $r_{crit} = 2.3\,m$, $r_{med} = 3\,m$.

Worker	Prediction					Ground Truth					RMSE	HNA (%)
	X	Y	Z	*d*	Risk	X	Y	Z	*d*	Risk		
0	4.82	-9.86	15.46	2.64	Caution	4.85	-10.34	14.29	2.31	Caution	1.26	90.97
1	6.15	-7.39	15.01	5.18	Safety	6.11	-8.44	14.29	4.33	Safety	1.27	90.90
2	9.30	-13.40	15.22	6.37	Safety	9.64	-14.64	14.29	7.13	Safety	1.59	88.67
Average											1.37	90.18

(b) Frame 398 with $r_{crit} = 2.51\,m$, $r_{med} = 3.26\,m$.

Worker	Prediction					Ground Truth					RMSE	HNA (%)
	X	Y	Z	*d*	Risk	X	Y	Z	*d*	Risk		
0	19.65	-46.72	14.77	16.46	Safety	16.90	-47.88	14.15	13.51	Safety	3.05	78.23
1	5.38	-51.28	14.50	1.64	Warning	6.11	-51.97	14.29	2.54	Warning	1.03	92.67
2	4.16	-52.43	14.63	1.48	Warning	4.85	-52.37	14.29	2.52	Warning	1.14	91.86
3	4.47	-64.02	13.93	13.04	Safety	5.47	-62.37	14.21	11.50	Safety	1.94	86.13
4	22.09	-45.29	10.55	19.20	Safety	18.53	-47.05	14.15	15.29	Safety	5.36	61.71
5	1.87	-65.70	14.16	14.82	Safety	3.32	-63.30	14.15	12.31	Safety	2.80	79.97
6	0.76	-65.95	13.64	15.25	Safety	2.19	-63.21	14.15	12.31	Safety	3.13	77.62
Average											2.64	81.17

Table 5
Comparison between depth handling strategies and geometry projection baseline for 3D worker localization. Distances are in meter.

Method	X_DEV	Y_DEV	Z_DEV	RMSE	HNA%	Calibration method
Oracle planar projection (fixed height)	0.51	0.52	0.09	0.83	94.07	Manual
Unscaled depth	5.62	8.14	2.97	12.22	49.03	Automatic
Scaled depth (Ours)	**0.98**	**1.22**	**1.07**	**1.77**	**87.39**	**Automatic**

results show that frames with fewer objects achieve HNA values of up to 95%, while in more crowded scenarios, the localization performance remains stable, with an average HNA of 81.17%. Notably, the HNA values of workers in medium and critical risk zones are consistently higher than 90%. These results confirm that the proposed pipeline maintains reliable 3D localization under crane-mounted camera configurations, ensuring that potential hazards can be effectively monitored with high confidence.

To evaluate the effect of depth scaling, we compare our 3D localization results with two baselines: an oracle planar projection and unscaled monocular depth as shown in Table 5. The oracle baseline estimates 3D worker positions by projecting 2D image points onto a planar ground surface using a manually calibrated projection matrix with fixed-height assumptions and frame-wise point selection, serving as an upper-bound reference. While this setting achieves the lowest localization error (RMSE of 0.83 m and HNA of 94.07%), it relies on manual calibration and predefined geometric parameters, making it impractical for crane-mounted cameras with continuous rotation and dynamically changing viewpoints. In contrast, unscaled monocular depth suffers from severe scale ambiguity. The proposed sparse-guided depth scaling incurs a moderate increase in localization error (RMSE of 1.77 m and HNA of 87.39%) but operates in a fully automatic manner without requiring manual calibration or predefined scene parameters, offering a practical trade-off between accuracy and applicability for dynamic camera configurations.

4.3.2. Sensitivity to camera pose perturbations

As stated in the previous Section 4.1.2, pose inaccuracies may propagate to the sparse reprojection depth and consequently affect both depth scale estimation and the final 3D localization results. To quantify this effect, we conducted a controlled perturbation experiment to analyze how pose errors influence depth scaling and localization accuracy. In this experiment, synthetic noise was progressively added to the predicted camera poses along the three translation axes and three rotation axes. For each axis, three perturbation magnitudes were introduced, and the resulting variations in localization accuracy and depth scale estimation were measured.

Table 6 shows that camera pose perturbations mainly affect the final 3D localization accuracy rather than the depth scale estimation. Across all perturbation types, the depth scale variation remains relatively small, generally around 2%–3%, indicating that the proposed scale calibration strategy is robust to moderate pose errors. In contrast, localization accuracy is more sensitive to pose perturbations, with the average increase in RMSE ranging from 14.2% to 20.9% depending on the perturbation direction. Among translation perturbations, errors along the X-axis produce the largest degradation, reaching a 33.66% RMSE increase at ± 0.1 m. This suggests that lateral camera displacement introduces stronger reprojection inconsistencies during scene coordinate estimation. For rotational perturbations, rotations around the Y-axis show the highest sensitivity, with localization errors increasing up to 25.67% at $\pm 3°$, whereas rotations around the X-axis produce the smallest impact on both localization accuracy and scale estimation.

Overall, these results indicate that although pose inaccuracies propagate through the reprojection-based depth estimation process, the resulting scale estimation remains stable while localization accuracy degrades in a controlled manner. This behavior demonstrates that the proposed framework exhibits graceful degradation under moderate camera pose perturbations.

4.3.3. Digital twin performance

Fig. 16 illustrates a sample safety report, showing how the 3D positions of workers and suspended loads are integrated into a BIM-based digital twin environment in NVIDIA Omniverse. Dynamic safety zones are visualized in real time, where the red zone marks the critical fall area beneath the load, the yellow zone represents the oscillation range, and the remaining area is classified as safe. The monitoring dashboard updates frame by frame, reporting the exact number of workers in each risk level. In addition to real-time visualization, the framework automatically generates a historical safety report that logs per-frame safety information, including timestamps, worker identifiers, estimated 3D positions, distances to hazardous zones, and associated risk labels. The note field contains advisory safety messages derived from the estimated risk levels, providing recommendations such as evacuation or caution alerts. These messages are designed to assist human decision-making and situational awareness, rather than to initiate automatic control or enforcement actions. Table 7 shows the per-worker risk assessment defined based on the distance from predicted positions and ground-truth values to the center of hazard zones using d_{crit} and d_{med} thresholds. The results show that the model correctly reports the risk level for most workers, while discrepancies occur for a minority of cases due to centimeter-level deviations between estimated and ground-truth values. Moreover, Table 8 summarizes the overall results across 3600 video frames, reporting the total number of correct predictions for safety and caution warning cases relative to the total number of worker instances. It can be observed that the model maintains reliable performance over 3600 frames when handling a large number of objects, achieving a confidence score of up to 94.19%.

Table 9 reports the end-to-end runtime breakdown of the proposed framework under operational conditions. The total inference latency per frame is approximately 154 ms, corresponding to an average throughput of 6.48 FPS. The runtime distribution indicates that monocular depth estimation and sparse-to-dense depth scaling account for the majority of the computational cost, while camera pose estimation and 3D localization with risk assessment incur relatively low overhead. Although the achieved frame rate is lower than conventional video processing rates, it is sufficient for crane safety monitoring, where hazardous events evolve at a slower temporal scale than frame-level visual changes. These results demonstrate that the proposed system can support near–real-time safety reporting while maintaining stable performance during continuous operation.

From the case study, the proposed framework demonstrates the ability to visualize 3D localization results and perform risk assessment using only a camera system. Moreover, NVIDIA Omniverse USD Composer enhances safety monitoring by providing a digital twin interface with integrated 3D risk visualization, real-time 2D detection, and safety reporting, offering a potential means for interaction between workers and crane operators.

5. Conclusions

This paper presented a framework termed D2S-MDE, which leverages camera relocalization and monocular depth estimation to enable accurate and real-time 3D safety monitoring in tower crane operations. The proposed method addressed a critical gap in existing safety management systems: the reliance on either costly multi-sensor infrastructures or static camera calibration that cannot adapt to the highly dynamic conditions of construction sites. By demonstrating that

Table 6
Sensitivity analysis of localization accuracy and scale estimation under camera pose perturbations.

Perturbation	Magnitude	Δ Localization error (RMSE %)	Δ Depth scale (%)
Translation X	±0.02 m	13.078	2.05
	±0.05 m	13.831	2.04
	±0.10 m	33.66	2.03
	Average	**20.191**	**2.037**
Translation Y	±0.02 m	13.19	2.05
	±0.05 m	21.27	2.07
	±0.10 m	18.30	2.07
	Average	**17.583**	**2.061**
Translation Z	±0.02 m	12.75	2.04
	±0.05 m	19.05	2.09
	±0.10 m	23.62	3.27
	Average	**18.474**	**2.468**
Rotation X	±0.5°	17.072	1.98
	±1°	10.482	2.09
	±3°	15.19	1.66
	Average	**14.248**	**1.908**
Rotation Y	±0.5°	18.189	2.08
	±1°	18.890	2.09
	±3°	25.67	3.31
	Average	**20.915**	**2.495**
Rotation Z	±0.5°	18.092	1.87
	±1°	18.291	2.17
	±3°	20.08	2.11
	Average	**18.820**	**2.049**

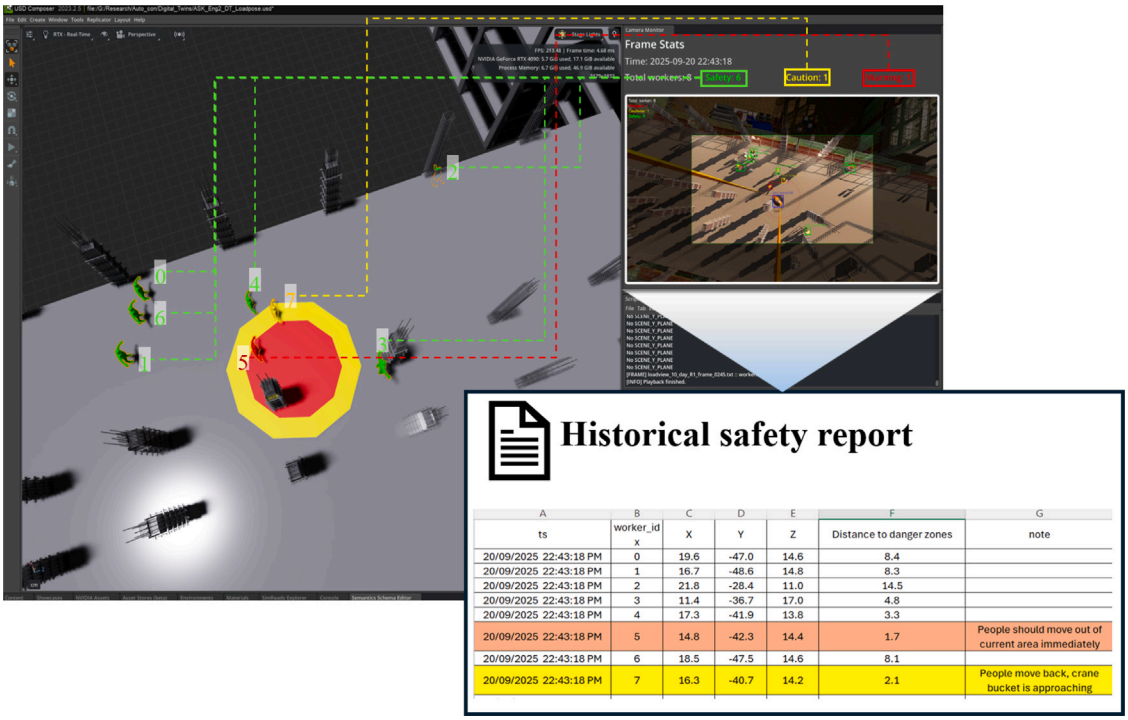


Fig. 16. Visualization of safety report for Frame 245.

reliable 3D localization can be achieved from a single monocular image, our framework provided a lightweight yet powerful alternative for vision-based safety monitoring. The contributions of this study were threefold. First, we proposed a calibration-free, single-image framework that combines scene coordinate regression and monocular depth estimation to recover metric 3D information under crane-mounted camera constraints. Second, we introduced a depth scale reconciliation strategy that aligns relative monocular depth with the metric scene coordinate space, ensuring geometric consistency for accurate 3D localization. Third, the digital twin was not introduced as a core methodological contribution or a closed-loop control component, but as a system-level integration and decision-support layer that operationalizes vision-based 3D localization into actionable safety assessment.

The proposed framework highlighted its ability to overcome key limitations identified in previous studies. Unlike traditional stereo vision or manual calibration approaches, the system maintained accuracy without requiring multiple viewpoints or extensive preprocessing. Compared to sensor-based methods, it avoided additional costs and

Table 7
Per-worker risk assessment for Frame 245: d_{crit} and d_{med} thresholds, predicted distance (m) with risk label, and ground-truth distance (m) with risk label.

Worker	d_{crit}	d_{med}	Prediction (m)	Risk Assessment	Ground Truth (m)	Risk Assessment
0	2.10	2.73	8.35	Safety	7.87	Safety
1	2.10	2.73	8.29	Safety	7.69	Safety
2	2.10	2.73	14.46	Safety	11.01	Safety
3	2.10	2.73	4.84	Safety	6.12	Safety
4	2.10	2.73	3.34	Safety	3.53	Safety
5	2.10	2.73	1.66	Warning	2.27	Warning
6	2.10	2.73	8.11	Safety	7.71	Safety
7	2.10	2.73	2.14	Caution	2.27	Warning

Table 8
Summary of correctly classified workers and overall risk classification accuracy across all frames.

Total frames	Total predictions	Correct predictions			Risk Classification Accuracy (%)
		Safety	Caution	Warning	
3600	20640	17600	1200	640	94.19

Table 9
End-to-end runtime breakdown of the proposed framework under operational conditions.
Performance on i9-13900KF CPU and RTX4090 GPU

Inference step	Time (s)	Share (%)
Camera pose estimation (D2S)	0.011	7.08
Monocular depth estimation	0.062	40.19
2D object detection	0.018	11.89
Sparse-to-dense depth scaling	0.036	23.33
3D localization and risk assessment within the digital twin runtime	0.027	17.51
End-to-end latency	**0.154**	**100.00**
Throughput (FPS)	**6.480**	

complexity by utilizing existing CCTV systems mounted on heavy machinery, while offering a lightweight and practical solution suitable for on-site deployment. Most importantly, by embedding the outputs into a digital twin, the framework not only localized workers but also continuously evaluated their safety status in relation to dynamic crane operations, which has not been sufficiently addressed in prior work. Overall, this study demonstrated that real-time 3D safety monitoring is feasible using a single crane-mounted camera, offering both academic and practical contributions. From an academic perspective, it advanced the state of the art by integrating scene coordinate regression and monocular depth estimation into a unified framework for safety-critical applications, enabling accurate assessment of overhead risks under dynamic camera conditions. From a practical perspective, it provided a scalable, lightweight, and deployable solution that supports proactive accident prevention in crane-intensive construction projects. The proposed framework represented a meaningful step toward the integration of computer vision and digital twin technologies for safer and smarter construction site management.

Despite the promising performance of the proposed framework, several limitations and future research directions should be acknowledged. Most notably, the proposed framework is sensitive to severe visibility degradation caused by fog, heavy rain, and low-light conditions. This limitation is inherent to monocular vision-based relocalization approaches and delineates the current operational boundary of the system. As demonstrated in the experimental results, the framework is primarily evaluated under clear or moderately favorable visual conditions, which correspond to the dominant operating regime for typical construction crane activities. In addition, the current risk assessment module relies on static geometric thresholds and does not explicitly model dynamic crane motion, wind effects, load mass, or temporal worker trajectories. While this formulation is sufficient for proof-of-concept visualization and early hazard indication, incorporating physics-based crane dynamics and spatio-temporal interaction modeling remains a challenging yet important direction for future research.

Future work will focus on improving robustness under adverse visibility and dynamic crane-camera conditions through uncertainty-aware relocalization and adaptive depth scaling strategies. Additional efforts will also investigate pose refinement and geometric consistency optimization to improve localization accuracy for distant targets and wide-angle crane-mounted views. Long-term field deployment during real crane operations will be conducted to evaluate practical reliability under varying weather, lighting, and worker-interaction conditions. Future work may also extend the digital twin from rule-based spatial risk assessment toward behavior-oriented safety management by incorporating simulation-based models of unsafe worker behavior and management intervention strategies [61]. Finally, future extensions may integrate Large Language Model (LLM)-based modules for natural-language safety reporting [62,63] and decision support within the digital twin environment [64].

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used GPT-5.0 in order to improve the readability and language. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

CRediT authorship contribution statement

Tran Dang Khoa Vo: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Yuntae Jeon:** Writing – review & editing, Data curation. **Tan-Dat Pham:** Writing – review & editing, Validation, Software, Data curation. **Pa Pa Win Aung:** Writing – review & editing, Visualization, Software. **Dongyoung Ko:** Writing – review & editing, Validation. **Seunghee Park:** Resources, Funding acquisition.

Declaration of competing interest

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Data availability

Data will be made available on request.

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